

MOHAMMED IRFAN K

PhD Student - Computer Vision, MBZUAI

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RESEARCH INTERESTS

Personalized Multimodal AI Assistants, Omni-Modal Large Language Models, Omni-Modal Agents and Tool Use, Omni-Modal Embedding and Retrieval Models, Benchmarks and Evaluation for Multimodal Reasoning, Real-Time Speech-to-Speech Conversational Systems, Multilingual and Multicultural Multimodal LLMs, Multimodal LLMs for Healthcare and Assistive Technology.

EXPERIENCE

March 2024
Present

PhD Student - Computer Vision (Aug 2025 - Present) | Research Associate II (Mar 2024 - Aug 2025) MBZUAI

- Contributed to multiple research papers, including **three EMNLP 2025**, **one ACL 2025**, and **two EMNLP 2026**, along with a **workshop challenge at CVPR 2025**.
- Received research recognition, including the **Meta Llama Impact Innovation Award 2024** for the **BiMediX2** project, **NVIDIA Academic Grant** for the ongoing extension of BiMediX2 with speech capability, and the **Meta Regional Research Grant** for our ongoing **OMER project**, which focuses on multilingual speech-visual LLMs.
- Built and presented **public demos** of our research at international exhibitions and events (see **Demos & Exhibitions**).
- **MediX-R1: Joint first author**, contributed to developing an open-ended **Reinforcement Learning** framework for medical MLLMs with composite reward signals enabling clinically grounded free-form reasoning beyond multiple-choice formats; accepted to **EMNLP Findings 2026**.
- **LongShOT: First author**, led the development of **LongShOTBench**, a diagnostic benchmark and agentic framework for omni-modal reasoning over long videos integrating vision, speech, and audio with tool use; currently **under review at NeurIPS 2026**.
- **Omni-Embed-Mini: First author**, led the development of a compact **omni-modal embedding model** that binds text, image, audio, and video into a shared space via **dense distillation**, retaining prior modality alignment without catastrophic forgetting; accepted to **EMNLP Findings 2026**.
- **BiMediX2: Joint first author**, contributed to **coding, idea development, benchmarking**, and **paper writing**. The paper has been accepted to **EMNLP Findings 2025**. The paper received wider media coverage such as **Yann LeCun on BiMediX2**
- **VIMUL**: Contributed to the multimodal **evaluation pipeline and toolkit development**. The paper has been accepted to **EMNLP 2025** and extended into a **workshop challenge at CVPR 2025**, in which the participants used our evaluation toolkit to benchmark their models.
- **LLMVoX: Second author**, contributed to the **idea formation**, leveraging my prior **speech domain experience**, and supported in **coding, brainstorming, and paper writing**. The paper has been accepted to **ACL Findings 2025** and was ranked as the **#3 Paper of the Day on Hugging Face** on March 7, 2025. Additionally, it received **100+ stars on GitHub** within the **first two days of release**.
- **MSc Student Mentoring**: Mentored Yevheniia Kryklyvets's **MSc thesis**, by actively engaging in **research discussions and ideation**, supporting and **assisting in model training**. This work, which involves nearly **1000 bird species** with **audio-visual analysis**, has been accepted to **EMNLP 2025**.
- **Mentored students** in the **UGRIP** internship program, **guiding and assisting** them through research and development projects at MBZUAI.

July 2022
February 2024

Project Associate | CEET-IIT Madras Research Park

- **Voice Banking**: Developed a multilingual **voice-command** based banking system with a Flask API for balance inquiry, transfers, and transaction history, combining **ASR, MT, TTS, and transliteration** across **seven Indian languages**. [\[Link\]](#)
- **Indian Sign Language Recognition**: Achieved **94%** accuracy on 100 gestures using **HMMs** and **89%** on 215 gestures with **Transformers**, using 3D pose extraction and normalization. Automated **dataset collection** from YouTube and built a **video annotation tool**.
- **Virtual Guide (AWCV Project)**: Built a **speech-controlled wheelchair** system in **seven Indian languages** with a custom **chatbot and real-time** question answering over web-scraped content. [\[Demo\]](#) [\[Project Page\]](#)

HONORS & AWARDS

- **NVIDIA Academic Grant Program Award** (2025): For the proposal "VILACare: Arabic-English Bilingual Speech-Visual LMM with Advanced Reasoning"
- **Meta Llama Impact Innovation Award** (2024): For "BiMediX2: Bio-Medical EXpert LMM for Diverse Medical Modalities"
- **Meta Regional Research Grant** (2024): For the proposal "OMER: Omni-Modal, Middle Eastern Culture-Specific, Ego-Centric Reasoning Dataset for Llama 3.2 Instruction Tuning"

PUBLICATIONS

Kurpath, M. I., Kaithakkodan, J. M., Mullappilly, S. S., Laptev, I., and Cholakkal, H. (2026). Omni-Embed-Mini: Binding modalities without forgetting via dense distillation: (to appear). In *Findings of the Association for Computational Linguistics: EMNLP 2026*

Kurpath*, M. I., Kaithakkodan*, J. M., Zhou, J., Mullappilly, S. S., Almansoori, M., Ahsan, N., Kalmakhanbet, B., Shikhar, S., Lalla, R., Lahoud, J., Awad, M., Khan, F. S., Khan, S., Anwer, R. M., and Cholakkal, H. (2025). A benchmark for omni-modal reasoning in long videos: (under review, NeurIPS 2026). *arXiv preprint arXiv:2512.16978* [\[Project Page\]](#)

Mullappilly*, S. S., **Kurpath*, M. I.**, Mohamed, O., Zidan, M., Khan, F., Khan, S., Anwer, R., and Cholakkal, H. (2026). MediX-R1: Open ended medical reinforcement learning: (to appear). In *Findings of the Association for Computational Linguistics: EMNLP 2026* [\[Project Page\]](#)

Shikhar, S., **Kurpath, M. I.**, Mullappilly, S. S., Lahoud, J., Khan, F. S., Anwer, R. M., Khan, S., and Cholakkal, H. (2025). LLMVoX: Autoregressive streaming text-to-speech model for any LLM. In *Findings of the Association for Computational Linguistics: ACL 2025* [\[Project Page\]](#)

Mullappilly*, S. S., **Kurpath*, M. I.**, Pieri, S., Alseiari, S. Y., Cholakkal, S., Aldahmani, K. M., Khan, F. S., Anwer, R. M., Khan, S., Baldwin, T., and Cholakkal, H. (2025). BiMediX2 : Bio-medical EXpert LMM for diverse medical modalities. In *Findings of the Association for Computational Linguistics: EMNLP 2025* [\[Code\]](#)

Shafique, B. S., Vayani, A., Maaz, M., Rasheed, H. A., Dissanayake, D., **Kurpath, M. I.**, Hmaiti, Y., Inoue, G., Lahoud, J., Rashid, M. S., Quasem, S. I., Fatima, M., Vidal, F., Maslych, M., More, K. P., Baliah, S., Watawana, H., Li, Y., Farestam, F., Schaller, L., Tymtsiv, R., Weber, S., Cholakkal, H., Laptev, I., Satoh, S., Felsberg, M., Shah, M., Khan, S., and Khan, F. S. (2025). A culturally-diverse multilingual multimodal video benchmark & model. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing* [\[Project Page\]](#)

Kryklyvets, Y., **Kurpath, M. I.**, Mullappilly, S. S., Zhou, J., Khan, F. S., Anwer, R. M., Khan, S., and Cholakkal, H. (2025). MAViS: A multimodal conversational assistant for avian species. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing* [\[Code\]](#)

Kurpath, M. I., Adwai, P. K., Bodireddy, J., K, C., and K, N. S. (2024). An IMUs and potentiometer-based controller for robotic arm-hand teleoperation. *Sensors and Actuators A: Physical*, 367 [\[DOI\]](#)

*Equal contribution / joint first authorship.

ACADEMIC SERVICE

- **Reviewer:** CVPR, NeurIPS, and Springer Nature journals.

EDUCATION

2025 Present	PhD, Computer Vision MBZUAI	GPA: 3.77/4
2021	B.Tech, Electrical and Electronics Engineering NIT Puducherry	CGPA: 8.66/10
2017	Senior Secondary Education (PCM) GMSSS Diglipur, Andaman & Nicobar Islands	Percentage: 85.6%
2015	Secondary Education GMSSS Diglipur, Andaman & Nicobar Islands	CGPA: 9.8/10

TECHNICAL SKILLS

- **Languages & Core Tools:** Python, Git, Linux
- **Deep Learning:** PyTorch, Hugging Face Transformers, OpenCV, vLLM
- **Training & Infrastructure:** LLM and multimodal LLM training and fine-tuning, distributed and multi-node training, SLURM, AWS HyperPod-class GPU clusters, NVIDIA and AMD clusters
- **Speech:** ASR, TTS, real-time speech-to-speech pipelines

- **Robotics & Edge:** ROS, Unitree GO2 quadruped, NVIDIA Jetson, Raspberry Pi, Arduino, ESP32, embedded sensors, IoT

DEMOS & EXHIBITIONS

- **Personalized wearable assistants for elderly care** with smartwatches, smart glasses, and non-invasive EEG devices, at [GITEX 2025](#) and **Machines Can Think (MCT) 2026**.
- **Multilingual multimodal LLM integrated with a robotic dog** (Unitree GO2) that holds **interactive conversations** and **carries out requested physical actions** while conversing, at [GITEX 2024](#), **Abu Dhabi Sustainability Week 2025**, and **MBZUAI Career Fair 2024**. [\[Demo\]](#)
- **BiMediX2 assistant** deployed over **Telegram**, presented at the **AI4SD event at UNGA79** and [GITEX 2024](#). [\[Demo\]](#)

OTHER PROJECTS

October 2024	Realtime Multimodal Conversational Model
GITEX Demo	A realtime multimodal conversational model with speech-in, speech-out over a lightweight VLM, enabling natural interactions. Integrated into the Unitree GO2 robot dog for seamless voice-based communication, and later extended into the personalized wearable assistants for elderly care demonstrated at GITEX 2025 and MCT 2026.
December 2022	Fun With Image Processing (Build Lab Course)
GitHub	An introductory course designed for beginners from diverse backgrounds, offered by the Build Club at IITMRP in collaboration with partner colleges. The course provides hands-on experience with OpenCV, featuring example problems, solutions, and warm-up exercises to assess understanding. Progress is tracked and shared through a dedicated Discord channel.
March 2022	Real-time Expression Recognition
GitHub Dataset	Developed a Real-time Expression Recognition model to identify facial expressions such as anger, disgust, fear, happiness, neutral, sadness, and surprise. Curated a dataset of 9,460 80x80 pixel color images from diverse age groups, with automated data cleaning and manual quality checks. Trained a CNN model, achieving 88.35% accuracy on the validation set and 82.58% on the test set. The model is deployed for real-time expression recognition via webcam and on videos.